1) Cancer Genomics Dataset

The cancer genomics dataset has 5479 features, each of which are different gene loci. I did not apply any preprocessing steps, but the dataset has already undergone feature selection based on variance to reduce its size, which is why certain gene loci are missing. The dataset has 801 samples, so we see that there are many more features than samples. The classes show some imbalance, with 300 BRCA samples, 146 KIRC samples, 141 LUAD samples, 136 PRAD samples, and 78 COAD samples. In addition, the feature values range between 0 and 21, with a standard deviation of about 3.5.

2) Model Performance (Heart Disease + Cancer Genomics)

Cross-Validation Results:

	Heart Disease		Cancer Genomics	
Metric	Mean	STD	Mean	STD
accuracy	0.9473170731707317	0.008392512455651324	0.986273291925466	0.006104854691187139
precision	0.9371628148666484	0.01152763540225626	0.991531772226752	0.00326352479973478
recall	0.9620125786163521	0.010236343677772862	0.9833128078817734	0.008229881945071774
f1	0.9493646193257081	0.007936233938412718	0.9870345257966233	0.005876334713557647
auc	0.9813686950366195	0.009396783108376706	0.9998028291878669	0.0002334600515504683

Based on these results, the heart disease dataset was harder to classify, with a lower mean and larger standard deviation for all performance metrics. The lowest metric for the heart disease dataset is precision, which means that false positives were more common. The lowest metric for the cancer genomics dataset is recall, which means that false negatives were more common.

3) Hyperparameter Tuning (Gradient Boosting)

For hyperparameter tuning, I used 3 cross-validation folds and tested the following values: max_depth = 3, 10, 20; n_estimators = 20, 50, 100; and learning_rate = 0.001, 0.01, 0.1.

For the heart disease dataset, the best parameters were max_depth = 10 or 20, n_estimators = 50, and learning_rate = 0.1. For the cancer genomics dataset, the best parameters were max_depth = 3, n_estimators = 50, learning_rate = 0.1.

Both datasets did better with a higher learning rate and a moderate number of estimators. However, deeper trees did better for the heart disease dataset, while shallower trees did better for

the cancer genomics dataset. This may have to do with the size of the cancer genomics dataset, which makes it more prone to overfitting.

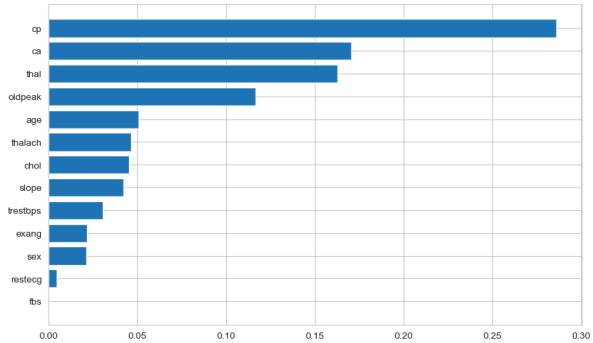
Grid Search Results:

Hypeparameters	Mean Test Score	
learning_rate, max_depth, n_estimators	Heart Disease	Cancer Genomics
0.001, 3, 20	0.896488	0.998849
0.001, 3, 50	0.896023	0.999419
0.001, 3, 100	0.900886	0.999219
0.001, 10, 20	0.988076	0.999648
0.001, 10, 50	0.983858	0.999525
0.001, 10, 100	0.984322	0.999564
0.001, 20, 20	0.988008	0.999648
0.001, 20, 50	0.983858	0.999525
0.001, 20, 100	0.984579	0.999564
0.01, 3, 20	0.906192	0.998929
0.01, 3, 50	0.923169	0.998929
0.01, 3, 100	0.937146	0.999638
0.01, 10, 20	0.981804	0.999327
0.01, 10, 50	0.983963	0.999746
0.01, 10, 100	0.986156	0.999805
0.01, 20, 20	0.981976	0.999321
0.01, 20, 50	0.984134	0.999746
0.01, 20, 100	0.986276	0.999805
0.1, 3, 20	0.951485	0.999033
0.1, 3, 50	0.975087	**0.999851**
0.1, 3, 100	0.982551	0.999847
0.1, 10, 20	0.985664	0.999739
0.1, 10, 50	**0.988701**	0.999793
0.1, 10, 100	0.985365	0.999796
0.1, 20, 20	0.983314	0.999739
0.1, 20, 50	**0.988701**	0.999793
0.1, 20, 100	0.985365	0.999796

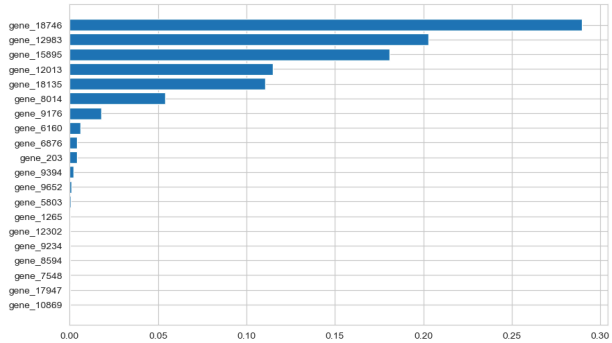
4) Feature Importance and Interpretation

<u>Gradient Boosting:</u>
For the heart disease dataset, the top 3 features by importance are cp, ca, and thal. For the cancer genomics dataset, the top 3 features are gene_18746, gene_12983, and gene_15895.

Heart Disease:



Cancer Genomics:



<u>HW1 Comparison:</u>
Based on the coefficients of the heart disease classifier (logistic regression) from HW 1, the top positive features are cp (1.00818678), thalach (0.47194734), and slope (0.38264811). The top negative features are sex (-0.86662727), ca (-0.83430974), and oldpeak (-0.71142362). Among these, cp, ca, and oldpeak are features that also had high importance during gradient boosting.

Below are the coefficients of a logistic regression model trained on the cancer genomics dataset, sorted by decreasing magnitude (absolute value) for BRCA, which is the most abundant class. The top gene loci vary for each cancer class, so we don't see much overlap between them.

```
['BRCA' 'COAD' 'KIRC' 'LUAD' 'PRAD']
         Feature         0         1         2         3         4
4337  gene_15589  0.017437 -0.003522 -0.003655 -0.005443 -0.004818
2351   gene_7964 -0.015460  0.003422  0.006521  0.004559  0.000959
4264  gene_15301 -0.015072  0.000669  0.003583  0.001006  0.009815
5380  gene_19661  0.013671 -0.001116 -0.002198 -0.001609 -0.008748
2919   gene_9652  0.013409 -0.003567 -0.004491 -0.004363 -0.000989
4935  gene_17905  0.012923 -0.002151 -0.002367 -0.004518 -0.003887
1413   gene_4773 -0.012635  0.000692  0.003597  0.000648  0.007697
2032   gene_6876  0.012116 -0.003691 -0.002289 -0.004022 -0.002114
117     gene_357  0.012035 -0.001050 -0.002845 -0.004608 -0.003532
2688   gene_8891 -0.012010  0.000671  0.002837  0.000670  0.007832
```

Overall, there is some overlap between the top features identified by both methods for the heart disease dataset, but there isn't really overlap for the cancer genomics dataset. Since we are using two methods that calculate feature importance differently, it makes sense that the important features don't necessarily agree, especially since the cancer genomics dataset has many features.

5) Comparison to HW1

Based on the gradient boosting feature importance (see plot above), only the top 13 features have importance, so I tested top-K feature selection for K = 3, 5, and 10.

Metric	Full	Top 10 Features	Top 5 Features	Top 3 Features
accuracy	1.0	0.9875776397515528	0.9875776397515528	0.937888198757764
precision	1.0	0.9870967741935484	0.9870967741935484	0.927077807250221
recall	1.0	0.9934426229508198	0.9934426229508198	0.9272836098826189
f1	1.0	0.99	0.99	0.9268855729761414
auc	1.0	1.0	0.9997387669801462	0.9935180609338824

Training a logistic regression model on a reduced feature set seemed to slightly hurt performance. Reducing from the full feature set to the top 10 or top 5 features only resulted in a small decrease in the metrics, but there is a larger decrease when reducing to the top 3 features.

6) Discussion

One limitation of the cancer genomics dataset is that there are many more features than samples, which could lead to overfitting. In addition, since some feature selection has already been done, we see that the cancer genomics dataset is easily classifiable, resulting in very high performance metrics. Both of these factors can make it difficult to distinguish between the effects of different hyperparameters. These limitations are also exacerbated by the fact that we are using a gradient boosting model, which is prone to overfitting and computationally expensive. Furthermore, because the cancer genomics dataset requires multi-class classification, we have to perform averaging of the performance metrics. This is impacted by class imbalances and can hide how performance varies between classes.